

Clever alloys

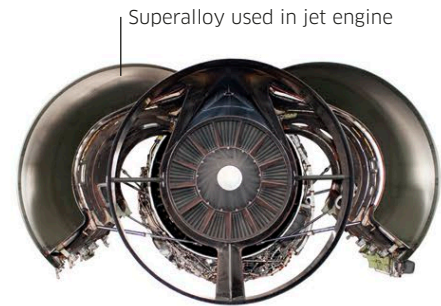
All alloys are developed to be an improvement on the individual metals from which they were made. Some alloys are an extreme improvement. Superalloys, for example, have incredible mechanical strength, resistance to corrosion, and can withstand extreme heat and pressure. These properties makes them very useful in aerospace engineering, as well as in the chemical industries. Memory alloys, or smart alloys, often containing nickel and titanium, “remember” their original shape.



With the help of just heat, this bent frame will snap back to its original shape.

Memory alloys

An object made from a memory alloy can return to its original shape if it has been bent. Simply applying heat restores the alloy to the shape it was in.



Superalloy used in jet engine

Superalloys

These high-performance alloys hold their shape in temperatures close to their high boiling points of around 1,832°F (1,000°C).

Spanish piece-of-eight

These legendary Spanish coins were made of silver. From the 15th to the 19th centuries, they were used throughout the vast Spanish Empire, and in other countries, too.

Japanese 50-yen coin

Copper (75%) and nickel (25%).



US dime (10-cent) coin

Copper (91.67%) and nickel (8.33%).

Swedish 10-krona coin

An alloy known as “Nordic gold,” also used in euro cents: copper (89%), aluminum (5%), zinc (5%), tin (1%).

Aluminum alloys

The metal aluminum is lightweight, resistant to corrosion, and has a high electrical conductivity. It is useful on a small scale (as foil, for example) but, because it is soft, it needs to be alloyed with other elements to be strong enough to build things. Aluminum alloys are often used in car bodies and bicycle frames.

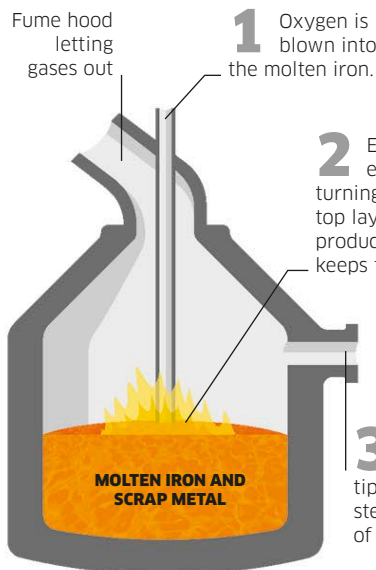
Light, rust-proof frame made of an aluminum alloy



Steel

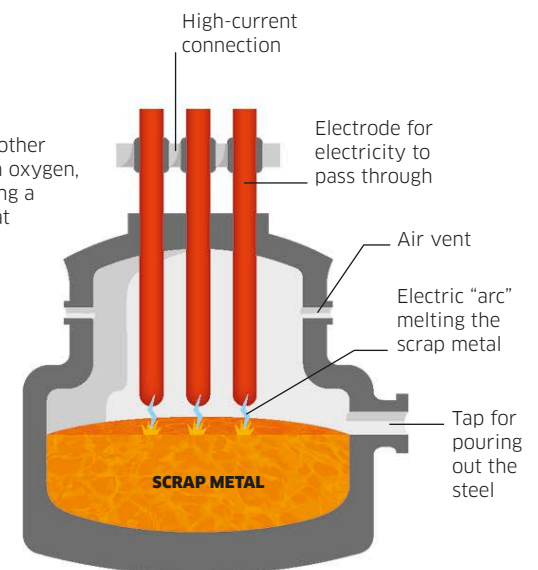
Iron, a pure metal, has been used since the Iron Age, some 3,000 years ago. But although it is very strong, iron is also brittle. There were some early iron alloys, but the strongest one, steel, came into common use during the Industrial Revolution in the 19th century. There are two ways of making steel: it can be produced from molten “pig iron” (from iron

ore) and scrap metal in a process called basic oxygen steelmaking (BOS), or from cold scrap metal in the electric arc furnace (EAF) process. Impurities, such as too much carbon, are removed, and elements such as manganese and nickel are added to produce different grades of steel. The molten steel is then shaped into bars or sheets ready to make into various products.



Basic oxygen steelmaking (BOS)

Oxygen is blown through molten “pig iron” and scrap metal to reduce its carbon content and other impurities. Then alloying elements are added, turning the molten metal into steel.



Electric arc furnace (EAF)

Cold scrap metal is loaded into the furnace. An electric current forms an “arc” (a continuous spark), which melts the metal. The final grade of steel is determined by adding alloying elements.